**Numerical Linear Algebra for Data Scientists** **Winter Semester 2026/27**

**Discussion:** 14 October 2026

# Exercise Sheet 1 (Lectures 1 and 2)

## Short-Answer Questions

**Exercise 1 (Rank and Column Space):**

The first two columns $A_1$ and $A_2$ of a matrix $A \in \mathbb{R}^{m\times 4}$ are linearly independent. The remaining columns satisfy

$$A_3 = 3A_1 - 2A_2, \qquad A_4 = -A_1 + 5A_2.$$

What is the rank of $A$? Does deleting $A_3$ and $A_4$ change the column space of $A$?

**Exercise 2 (Row Operations):**

Decide whether each of the following statements is true or false. No explanation is necessary.

a) Elementary row operations on an extended matrix always preserve the solution set of the corresponding linear system.
b) Elementary row operations always preserve the column space of a matrix.
c) Elementary row operations always preserve the rank of a matrix.

## Detailed Questions on Lecture 1

**Exercise 3 (Column Space):**

Find a matrix $A \in \mathbb{R}^{2\times 5}$ such that its column space $\mathcal{C}(A)$ is a line through the origin.

**Exercise 4 (Linear Dependence 1):**

Show: Two vectors in $\mathbb{R}^n$ are linearly dependent if and only if one is a scalar multiple of the other.

**Exercise 5 (Linear Dependence 2):**

Let $A \in \mathbb{R}^{m\times n}$. Suppose $A$ contains $n$ rows with exactly one non-zero entry each, and these entries appear in pairwise distinct columns. Show that the columns of $A$ are linearly independent.

**Exercise 6 (Identity Matrix):**

The identity matrix $I_n$ is the $n \times n$ matrix whose entries on the main diagonal are 1 and whose remaining entries are 0. Show that $I_n v = v$ for every vector $v \in \mathbb{R}^n$.

## Detailed Questions on Lecture 2

**Exercise 7 (Row Equivalence):**

Consider two row-equivalent matrices $A \sim B$ for $A, B \in \mathbb{R}^{m \times n}$. Show that if the $i$-th column of $A$ is zero, then the $i$-th column of $B$ is also zero.

**Exercise 8 (Gaussian Elimination):**

Use Gaussian elimination to solve the following system of linear equations. Indicate each elementary row operation and transform the extended matrix into row-echelon form before using back substitution.

$$\begin{aligned} x + y + z &= 6, \\ 2x - y + z &= 3, \\ x + 2y - z &= 2. \end{aligned}$$

**Exercise 9 (From Row-Echelon Form to Reduced Row-Echelon Form):**

Describe how a matrix in row-echelon form can be transformed into reduced row-echelon form using elementary row operations.

**Exercise 10 (Rank and Row-Echelon Form):**

We have seen that the rank of a matrix $A \in \mathbb{R}^{m \times n}$ equals the number of pivots in its reduced row-echelon form. Does the same hold for any row-echelon form of $A$?

**Exercise 11 (Uniqueness of the Row-Echelon Form):**

Show that a matrix may have more than one row-echelon form, whereas its reduced row-echelon form is unique. For which matrices is the row-echelon form unique?